

A Review of the Concept of Impulsivity

An Evolutionary Perspective

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Impulsivity is a central concept for many psychological models and interventions. Despite extensive research, the concept of impulsivity lacks clarity due to the diversity of definitions provided. The aims of this study were to conduct a concept analysis of impulsivity to provide a clear and workable definition of impulsivity for future nursing research. This study used Rodgers' evolutionary concept analysis approach to analyze 37 articles from various disciplines. Our analysis offers promising direction in refining the concept. The findings can drive future research and enhance nurse researchers' ability to attend and explore health-related issues associated with the concept. **Key words:** *evolutionary concept analysis, impulsivity, nursing, Rodgers' evolutionary framework*

IMPULSIVITY is a relatively old concept that is receiving increasing attention in the literature. Between the late 70s and late 90s, there were about 650 articles in the Social Science Citation Index that contain “impulsivity” in the title,¹ and by the time of writing their book, Webster and Jackson¹ found over 700 web page matches the term impulsivity on the World Wide Web. In a previous literature search on the concept of impulsivity using the PsycINFO database between the years 1960 and 2012, we found 1805 arti-

cles associated with the concept. While there were only about 400 articles published between the years 1960 and 2000, more than 1400 articles were published between 2000 and 2010 (Figure 1); the scallop pattern of the accumulative number of publications about impulsivity indicates there is increased interest in impulsivity in the literature. However, a current search using the Cumulative Index of Nursing and Allied Health Literature (CINAHL) yielded no articles about impulsivity in nursing.² This search showed that the concept of impulsivity did not receive a comparable interest as in other fields of research. Although, as it has been described later, the concept was associated with various human problematic behaviors.

Impulsivity is associated with a wide range of human behaviors. For example, impulsivity has been used to predict a wide range of maladaptive and problematic behaviors such as gambling,³ hazardous drinking,⁴ overeating,⁵ addictive behaviors,⁶ and self-care behavior.⁷ Although impulsivity seems to be associated with health-related issues and behaviors, and despite the increasing interest in impulsivity in many fields, it appears that nursing researchers did not conduct specific studies

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Statement of Significance

What is known or assumed to be true about this topic?

Impulsivity is a central concept in various psychological models and interventions that was extensively investigated and supported in the literature. We found that the literature introduces various definitions for impulsivity, but these definitions lack the agreement on important aspects of such important behavioral and health-related phenomena.

What this article adds:

This is the first article to investigate the concept of impulsivity. It provides nursing researchers with workable definition of impulsivity. Attributes, antecedents, consequences, and related terms highlight that impulsivity a multidimensional concept that requires further investigation to guide further clarification of the concept and its potentials in nursing.

related to health-related human behaviors. Thus, the concept has been introduced into nursing literature as a potential researchable variable in nursing.

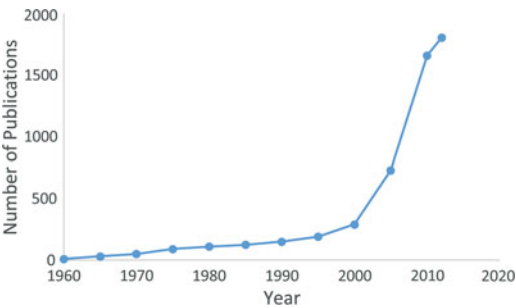


Figure 1. The accumulative number of articles about impulsivity retrieved from the PsycINFO database. From AL-Hammouri and Hall.² Adapted with permission. This figure is available in color online (www.advancesinnursingscience.com).

One potential problem of adopting the concept could be the lack of agreement on one unifying definition of impulsivity. Various disciplines introduced different definitions for the concept (Table). One explanation for the diversity of the definitions is the complexity and heterogeneity of impulsivity as a concept.¹⁴ These definitions provide different perspectives regarding the conceptualization of impulsivity in different fields. For example, in the neuroimaging science field, it seems that impulsivity is viewed as a characteristic of making decisions about single discrete behaviors. On the other hand, in the natural sciences it is viewed as a characteristic of determining how individuals

Table. Examples of Diverse Definitions of Impulsivity

Article	Field of Study	Definition of Impulsivity
Lv et al ⁸	Neuroimaging	Choosing an immediate smaller reinforcement over a larger delayed one
Weizmann-Henelius, et al ⁹	Psychiatry-social science	A tendency to act without planning or thinking
Khurana and Romer ¹⁰	Psychology	A multidimensional concept that is characterized by diminished control over behavioral urges
Wegmann et al ¹¹	Natural and clinical sciences	Impaired inhibitory control
Shorey et al ¹²	Interpersonal violence	A rapid decision-making characterized by carelessness
Shead et al ¹³	Mental health	Thoughtless acting

make decisions about a group of behaviors. These variations of the concept may lead to confusion and misunderstanding of what constitutes the concept of impulsivity or impulsive behavior. Thus, there is a need to analyze the concept of impulsivity to develop a homogeneous definition of impulsivity to easily delineate its basic attributes for a better understanding of the concept and corresponding behaviors it describes. Besides, we believe that a homogeneous definition of impulsivity will facilitate adopting the concept in nursing research and knowledge development in this area. Thus, this article's goals were to conduct a concept analysis of impulsivity to provide a greater understanding of this phenomenon and introduce a clear, comprehensive, and workable definition of impulsivity for future nursing research using Rodgers' evolutionary framework.

METHODS

The evolutionary approach is an inductive approach to concept analysis.¹⁵⁻¹⁷ The focus of this approach is on clarifying the concept and its current use to promote further investigation and development. This approach does not specify what the concept is or what is not; the goal is to provide a base for further research and development of the concept analysis cycle while being attentive to the development of new knowledge about the concept.^{15,18} A well-defined concept has potential descriptive and explanatory capabilities that may promote knowledge development in nursing.¹⁹ The evolutionary approach is a rigorous scientific approach to concept analysis, especially when the analysis is done across various disciplines and literature, which is the case of impulsivity in this review. By employing the evolutionary approach to analyze impulsivity, we hope to provide a working and precise definition of the concept of impulsivity that is amenable for further future development and to introduce impulsivity as a researchable phenomenon concerning health-related

behaviors of interest to nursing. The concept analysis conducted in this article went through 3 phases, as discussed next.

Phase 1: Literature search and article selection

The initial search was done using a local university library search engine in the following databases: CINAHL, PsycINFO, and EBSCO Host. We selected articles that met the following inclusion criteria: (1) studied human subjects; (2) explicitly stated a definition for impulsivity or provided a detailed example of impulsive behavior; (3) written in English; and (4) full-text articles. The keywords used in the search were impulsivity, impulse control, and impulsive behavior. The initial search yielded thousands of articles. Thus, the results were refined based on the following criteria: (a) peer-reviewed articles, (b) articles published since 2000, (c) written in English, (d) full-text articles, and (e) articles reporting findings of studies conducted among human participants. A total of 872 articles were retrieved and deemed appropriate for the current review of the concept of impulsivity. After excluding duplicate and irrelevant articles based on the title and abstract, we ended up with a total of 37 articles for the current analysis (Figure 2).

Phase 2: Extracting definitions, relevant examples, and measures

In this phase, we reviewed the relevant articles and extracted definitions, examples, and impulsivity measures with a focus on attributes, antecedents, consequences, and related concepts of impulsivity. Also, a separate literature search was done to look for related concepts that might be discussed concerning impulsivity to make sure that all surrogate concepts are included in the current analysis. Notes during this in-depth review of the selected articles were also considered during phase 3. This was done by the first 2 authors in this study separately to make sure the comprehensiveness of the reviews. The data were

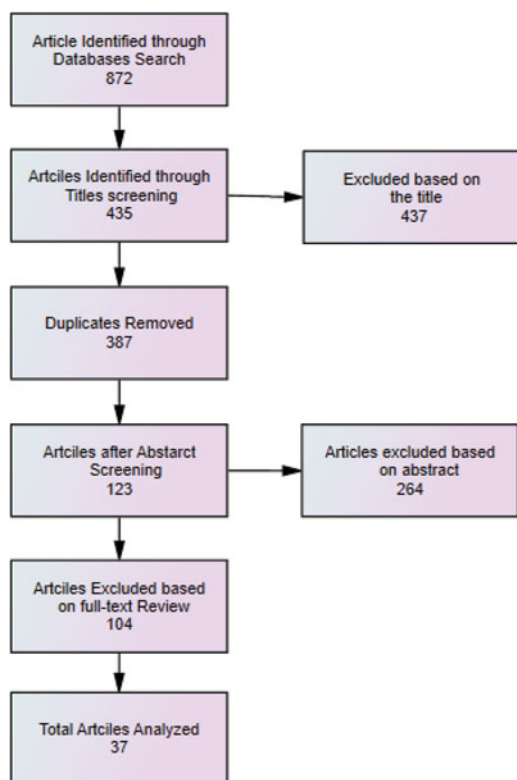


Figure 2. Literature search results. This figure is available in color online (www.advancesinnursingscience.com).

then compiled into 1 document to be entered into the NVIVO for analysis.

Phase 3: Thematic analysis

In this phase, the data extracted from phase 3 were entered into NVIVO 12 for thematic analysis. Two researchers separately analyzed the data to delineate attributes, antecedents, consequences, and related concepts of impulsivity. Then, a qualitative thematic analysis using NVIVO 12 software was used to analyze the data. The researchers then met and discussed areas of disagreements in coding until they reached 100% agreement, the same way it is done in thematic analysis in qualitative analysis. Finally, the results were reviewed by all 3 authors of the study to ensure experts' validity of the current analysis results.

RESULTS

The findings of the thematic analysis of impulsivity are presented in this section, and headings reflective of the evolutionary concept analysis approach.

Attributes of impulsivity

Attributes are the characteristics that are most frequently associated with the concept. Despite the heterogeneity in defining impulsivity, the current analysis provided us with some common attributes found in different definitions, examples, and impulsivity measures. The current analysis showed that attributes of impulsivity were best described according to 5 themes: (1) a characteristic of decision-making behavior; (2) free will; (3) insufficient reasoning; (4) individualized; and (5) normative.

Impulsivity as a characteristic of decision-making behavior

Most discussions about impulsivity presented it as a characteristic of the decision-making process to make choices among different behavioral alternatives. Although it has been shown that impulsivity is affected by environmental and situational variables,²⁰ impulsivity sometimes is described as a trait.²¹ In other words, an individual is expected to act the same way, in terms of impulsivity, under relatively stable environmental conditions. On the other hand, when the environmental condition changes, the level of impulsivity may change accordingly. For example, if a person with heart failure is consistently choosing not to follow his prescribed medications, we would reasonably expect that person with heart failure may not follow the prescribed regimen for exercises or diet, given that different environmental conditions are controlling it. Thus, impulsivity is better described as a characteristic of decision-making behavior rather than defining subjects' personality. In other words, we may need to assess different behaviors under different conditions rather than merely

labeling the person to be impulsive for a few instances of impulsive behaviors under certain conditions. The attribute of impulsivity as a characteristic of decision-making behavior will help researchers capture the environmental factors affecting impulsive behaviors.

Free will to make choices

The free will can be seen as a deliberate act of choosing, with at least 2 nonrandom options must be available.²² As discussed previously, impulsivity is a characteristic of subjects' choice-making behaviors that involve at least 2 options with different immediate and delayed outcomes. In other words, the person must pick one alternative among 2 or more available options freely. Thus, to label the choice of a specific behavior as impulsive, the individual must be free to choose between 2 or more alternatives (ie, nonrandom options). Under all definitions of impulsivity, the decision must be made based on an individual's analysis of each behavioral alternative's expected outcomes and the momentary value assigned for each option. For example, expecting a physically debilitated person to take medication independently would not be an example of impulsive choice because the free will does not apply in this case.

Insufficient reasoning

The reviewed articles described different forms of lacking reasoning. Some of the discussions indicated a lack of appropriate input, such as lack of attention and foresight.^{23,24} Others referred to a problem within the reasoning process itself, such as discounting the value of the nonimpulsive choices, as they lie far into the future.²⁵ Further, others tried to characterize the reasoning process, in general, to be lacking sufficient processing, such as lack of planning,²³ thoughtlessness acting on the spur of the moment,²⁶ and restlessness.²⁷ Regardless of the reason for lacking appropriate reasoning, the result of such lacking is an impulsive choice.

Thus, another attribute of impulsivity is the diminished reasoning process. Making choices among different alternatives requires weighing all possible alternatives thoroughly. In general, we expect the person to pick the most appropriate course of action that leads to the best outcomes or minimize potential losses. For example, the impulsivity level in children with attention-deficit hyperactivity disorder (ADHD) is higher than typically developed children.²⁸ This is expected given that ADHD and intellectual disabilities, which affect logical reasoning, are prevalent co-occurring problems.²⁹ Thus, impulsivity under some proposed definitions is described as acting on the spur of the moment without considering future consequences of the behavior (see the Table).

Individualized

Another attribute of impulsivity is that it is different among different individuals. Similar to other concepts, impulsivity varies among individuals. Such variation has been addressed by creating various tools to measure impulsivity, with the Barratt Impulsiveness Scale (BIS-11) being the most widely used measure. The BIS-11 is a self-report scale with Likert-type scale answer options.³⁰ The items of the BIS-11 ask about a person's tendency to act in a way that does not match the objective value of specific behaviors listed in the scale. These items are rated in Likert-type scale response options ranging from 1 (rarely/never) to 4 (always). The BIS-11 determines the level of impulsivity of the individual by calculating the total score. A score of 72 or above indicates high impulsivity, a score of 52 to 71 indicates normal impulsivity, and a score below 51 indicates overcontrol. Besides, there are variations in the impulsivity levels that may be present not only among different individuals but even among different cultures.³¹

Normative

The label for this attribute was borrowed from Cowles and Rodgers³² in their analysis

of grief.¹⁵ Impulsivity is not always a negative behavioral trend; sometimes, it may be essential and advantageous. For example, for a hungry person lost in a desert, waiting to get healthy food may lead to his death, while eating what is readily available may sustain his life. In this regard, impulsivity can be considered part of our psychological being, but the problem lies in its level. Normative, in this context, means that, despite the individual differences, there are always acceptable levels of impulsivity regardless of its variability. Some authors specified the appropriate, normal, or acceptable ranges and forms of impulsivity.^{30,32} For example, using the BIS-11 to measure impulsivity, Stanford and colleagues³⁰ suggested categorization based on the BIS-11 scores as discussed previously. This would mean that impulsivity is an inherent part of human psychology, but there is an optimum level to function in a given environment appropriately.

Multidimensional

Although multidimensionality was mentioned in the definition of impulsivity, this theme was extracted in the current analysis from the measures used to assess impulsivity. The most widely used measure to evaluate impulsivity is the BIS-11.³⁰ The BIS-11 assesses 3 types of impulsivity: motor impulsiveness, nonplanning, and attentional impulsiveness. The next most widely used measure is the UPPS.³³ The UPPS assesses the 4 types of impulsivity: urgency, premeditation, perseverance, and sensation seeking. Despite the difference in conceptualizing the multidimensionality of impulsivity across the most widely used impulsivity measures, they all agree on the multidimensional nature of the concept.^{30,33} In other words, an individual may tend to be more impulsive in some behavioral domains than others, which can be considered a source of personal differences among different individuals.

Antecedents

Rodgers stressed the importance of determining the events that cause the concepts to occur.¹⁸ We found 4 antecedents of impulsivity: choice characteristics, developmental history, culture, and situational factors. Choice characteristic is a theme that consists of 3 components as an antecedent for impulsivity. First, the presentation of choice is between 2 or more alternatives.¹⁶ Second, the choice alternatives must be significantly different in value for the person making the choice in that at least one with favorable outcomes and another with less favorable outcomes.⁸ The third characteristic is the temporal assembly of the choices; the favorable choice outcomes must be far in the future, while the outcomes with less favorable choice must be more immediate.⁸

Developmental history is another antecedent of impulsivity. For example, individuals who suffered childhood abuse were found to be more impulsive as adults compared with those who did not suffer childhood abuse.³⁴ Another example is that individuals who developed in a high-risk environment also have higher impulsivity levels.³⁵ Another theme that emerged as an antecedent for impulsivity is culture. For example, the Americans and Chinese discounted delayed rewards more steeply than the Japanese.³¹

The last theme that emerged as an antecedent for impulsivity is situational factors. We used situational factors to refer to the environmental factors present at the time of making a choice but not part of the choice assembly itself. These factors include variables such as stress, hunger, and deprivation. For example, individuals who suffered hunger during making choices are found to be impulsive even if the choice made was not associated with access to food, and the same applies to the deprivation.³⁶ For stress, it has been suggesting that if the individual making a choice was suffered from stress, the individual tends to be more impulsive.³⁷

Consequences

According to Rodgers,¹⁸ the consequence is the event that follows the occurrence of the concept. We found 2 themes that emerged as consequences of impulsivity: increased risk of adverse outcomes and emotional impact. We discovered that impulsive choice will lead to poor self-care behavior in persons with heart failure that, in turn, will lead to a progression in the disease process and further adverse outcomes.⁷ Daniel et al³⁸ found that impulsivity related to monetary gain was associated with the lack of success to weight loss in obese individuals. Another example is the study's findings by Ryu et al³⁹ that showed that increased impulsivity associated with behaviors such as gambling, sexuality, and eating led to lower quality of life in individuals with Parkinson disease.

The second theme, emotional suffering, can be manifested in experiencing 2 types of emotions. The literature showed that some impulsive choices yield negative emotions such as regret,⁴⁰ while others impulsive behaviors generated positive emotions such as joy and satisfaction.⁴¹ However, those 2 studies investigated the effect of emotional postpurchase outcomes. The difference in the resulted emotions could be a result of the reason causing impulse purchase. The positive emotions in postpurchase emotional impact may lie in that persons who experienced such positive emotions engaged in impulse purchasing stress relief behavior. Thus, further investigation is required in this area for more robust evidence on emotional impact post-impulsive behavioral choice controlling for the procedural difference between conflicting studies.

Definition of impulsivity

Based on the current analysis and the previously mentioned attributes, impulsivity can be defined as an individualized, normative, and multidimensional pattern of human decision-making behavior characterized by free will and insufficient reasoning due to the

diminished reasoning process. This definition is believed to encompass all the definitions and examples of impulsivity found in the literature.

Related concepts

The concepts of impulsivity in the literature are delay discounting,⁴² self-control,⁴³ and delayed gratification.⁴⁴ Delay discounting is seen as the underlying process of impulsivity.⁴² Delay discounting describes how human subjects discount the larger rewards' value as the time between making a choice and access to the larger reinforcer increases.⁴² It is expressed in the form of a curve, and the slope of the curve represents the speed we discount the value of the larger rewards concerning the time. Although the shape of discounting rate with time unit is almost the same for all human subjects, the slope is not⁴²; it means that we all discount future rewards in a way or another, but the speed in which we are doing that is different. Theoretically, we may be able to change the speed of discounting. Still, we will not be able to omit it from our behavioral repertoire as a process that affects our behaviors.

On the other hand, some research on impulsivity suggested some acceptable levels of impulsivity,³⁰ but when we research those levels, we start calling associated behaviors or individuals as self-controlled.³⁰ This moves us to the self-control concept, which, based on the literature, was introduced as one end of impulsivity when impulsivity at its lower levels.³⁰ Although the question, if the underlying process(s) behind impulsivity and self-control is the same or different, is not answered yet,⁴³ the literature is centered on interventions targeting impulsivity rather than promoting self-control.⁴³ For delayed gratification, it is less frequently used than impulsivity, but it was used interchangeably with impulsivity.⁴⁴ The ability to delay gratifications has been used to describe the lower levels of impulsivity, while the inability to delay gratifications has been used to describe

higher levels of impulsivity. Still, no clear measures or interventions deal with the concept of delayed gratification other than those targeting impulsivity.

Identifying an exemplar

Providing an exemplar, according to Rodgers,¹⁸ provides a practical demonstration of the concept in the field of interest. The exemplar should be extracted from the real world rather than being constructed or invented. The exemplar can be derived from one's own or others' personal experiences. In a secondary health care center, the first author of this article assigned students to teach patients to meet this requirement. One of the patients assigned to the students was Sarah. Sarah, a pseudonym for the actual patient, suffered from heart failure, type 2 diabetes, and a body mass index of more than 30. Sarah's doctor prescribed a specific plan to minimize the exacerbations of her conditions. Sarah's plan involved a weight reduction component. Sarah's impulsivity score on the BIS-11 was 95, indicating high impulsivity (attribute: normative). The assessment of Sarah's dietary habits revealed that obesity primarily originated from her dietary choices (attribute: impulsivity as a characteristic of decision-making behavior). Sarah's food preferences included such food choices that are high-calorie, high-fat, and salty. Sarah had received teaching classes about her dietary plan and provided the knowledge to decide on what types of food she had to eat to minimize her disease progression. Sarah's behavior emerged every time she was confronted with alternative food choices picked personally by her (attribute: free will to make choices). Sarah did not fully consider the future consequences of her behaviors (attribute: insufficient reasoning). Usually, after eating a high-calorie meal, Sarah regretted her choice and promised herself not to do it again. However, when she was confronted with the same situation again, she picked the same choices every time. Sarah continued gaining weight. In her previous attempts,

Sarah was able to follow her exercising schedule given by health care professionals, but she could not follow the dietary recommendations (attribute: multidimensional).

In contrast, Sarah's friend, who had a BIS-11 score of 82, was more successful in following his dietary and exercise regimen. Sarah's friend was able to drop his body mass index from 29 to 26 within a year (attribute: individualized). Reviewing the medical history of Sarah, we identified many sources of stress in her life, such as her heart failure and diabetes conditions and her role as a primary caregiver for her 3 children. Stress was identified as antecedents for her impulsive choices. Her grief and worsening of her heart condition were identified as consequences of her impulsive choices.

DISCUSSION

Impulsivity is a widely researched concept in various fields. However, there is no concrete conclusion regarding its definition.⁴¹ Since the concept was defined and used by many varied fields, we used Rodgers' evolutionary framework to conduct a concept analysis of impulsivity to provide a greater understanding of this phenomenon and introduce a clear, comprehensive, and workable definition of impulsivity for future research in nursing and other fields of study. The definitions of impulsivity are numerous and diverse. The current analysis yields a definition for impulsivity that is amenable to further research and development.

The current analysis added to the concept of further insight by introducing antecedents and consequences of impulsivity. We also introduced related concepts that are usually discussed with impulsivity and share some of their attributes. As we discussed earlier, these concepts were presented as processes that underlie impulsivity or concepts that share impulsivity attributes but were not researched extensively as impulsivity. We also added a model case that explains how attributes, antecedents, and consequences can

be applied to analyze any impulsive behavior in the light of the current analysis.

The importance of adopting impulsivity into nursing research lies in that impulsivity is a behavioral operant; the term used in behavior analysis to describe behaviors that can be manipulated, increased or decreased, through the teaching/learning process, by manipulating the environment.⁴⁵ Several studies described specific interventions that can effectively reduce impulsivity, such as framing, priming, reward bundling, brain training, and contingency management.⁴⁶⁻⁴⁸ Thus, whenever impulsivity turned to be an effective contributor in any nursing related phenomenon, a whole new interventional research area will open to check the effectiveness of these interventions to deal with the phenomenon of interest. As discussed earlier, impulsivity is a predictor of various problematic human behaviors that can jeopardize persons' health outcomes with or at risk for chronic health problems. Integrating impulsivity into nursing research that focuses on such central concepts as self-care

and health promotion seems promising and is warranted. Currently, we are working on testing the potential of impulsivity and its role in self-care in persons with heart failure, and the results are promising and show that impulsivity can be a significant predictor of self-care in persons with heart failure.⁷

CONCLUSION

Some authors argued that impulsivity is a very complex concept in that there is no one single definition that can encompass the concept of impulsivity.⁴⁹ In the literature, there is clear evidence of differences in the conceptualization of impulsivity. Thus, the concept's current analysis was needed to provide a clear and user-friendly definition of such a complicated concept. We believe that the present analysis provided a working definition that encompasses all of the attributes of the concept that will hopefully help introduce and adopt the concept of impulsivity into nursing literature.

REFERENCES

1. Webster CD, Jackson MA, eds. *Impulsivity: Theory, Assessment, and Treatment*. Guilford Press; 1997: 462.
2. Al-Hammouri MM, Hall LA. A critical review of the state of measurement of impulsivity. Poster presented at the 27th Annual Conference of the Southern Nursing Research Society, Little Rock, AR, February, 2013.
3. Ioannidis K, Hook R, Wickham K, Grant JE, Chamberlain SR. Impulsivity in gambling disorder and problem gambling: a meta-analysis. *Neuropsychopharmacol.* 2019;44(8):54-61. doi:10.1038/s41386-019-0393-9
4. Kaltenegger HC, Låftman SB, Wennberg P. Impulsivity, risk gambling, and heavy episodic drinking among adolescents: a moderator analysis of psychological health. *Addict Behav Rep.* 2019;10:100211. doi:10.1016/j.abrep.2019.100211
5. Adams RC, Sedgmond J, Maizey L, Chambers CD, Lawrence NS. Food addiction: implications for the diagnosis and treatment of overeating. *Nutrients.* 2019;11(9):2086. doi:10.3390/nu11092086
6. Fowler J, Gullo MJ, Elphinstone RA. Impulsivity traits and Facebook addiction in young people and the potential mediating role of coping styles. *Pers Individ Dif.* 2020;161:109965. doi:10.1016/j.paid.2020.109965
7. Al-Hammouri MM, Rababah JA, Hall LA, et al. Self-care behavior: a new insight of the role of impulsivity into decision making process in persons with heart failure. *BMC Cardiovasc Disor.* 2020;20(1): 349. doi:10.1186/s12872-020-01617-8
8. Lv C, Wang Q, Chen C, Xue G, He Q. Activation patterns of the dorsal medial prefrontal cortex and frontal pole predict individual differences in decision impulsivity. *Brain Imaging Behav.* 2021;5(1):421-429. doi:10.1007/s11682-020-00270-1
9. Weizmann-Henelius G, Putkonen H, Rissanen T, Eronen M, Webster CD. Exploring a new structured professional judgment measure (impulsivity measure related to violence) after an average follow-up of 10 years: a study of Finnish offenders. *Crim Behav Ment Health.* 2019;29(1):57-68. doi:10.1002/cbm.2107
10. Khurana A, Romer D. Impulsivity in adolescence: predictors and consequences. *Encycl Child Adolesc Dev.* 2020. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781119171492.wecad472> Accessed June 15, 2021.

11. Wegmann E, Müller SM, Turel O, Brand M. Interactions of impulsivity, general executive functions, and specific inhibitory control explain symptoms of social-networks-use disorder: an experimental study. *Sci Rep*. 2020. <https://www.nature.com/articles/s41598-020-60819-4>. Accessed May 17, 2021.
12. Shorey RC, Brasfield H, Febres J, Stuart GL. The association between impulsivity, trait anger, and the perpetration of intimate partner and general violence among women arrested for domestic violence. *J Interpers Violence*. 2011;26(13):2681-2697. doi: 10.1177/0886260510388289
13. Shead NW, Champod AS, MacDonald A. Effect of a brief meditation intervention on gambling cravings and rates of delay discounting. *Int J Ment Health Addict*. 2020;18(5):1247-1263. doi: 10.1007/s11469-019-00133-x
14. Davis-Reyes BD, Campbell VM, Land MA, Chapman HL, Stafford SJ, Anastasio NC. Profile of cortical N-methyl-D-aspartate receptor subunit expression associates with inherent motor impulsivity in rats. *Biochem Pharmacol*. 2019;168:204-213. doi:10.1016/j.bcp.2019.07.007
15. Cowles KV, Rodgers BL. The concept of grief: an evolutionary perspective. In: Rodgers BL, Knafl KA, eds. *Concept Development in Nursing: Foundations, Techniques, and Applications*. W. B. Saunders; 2000:103-117.
16. Miller HN, Thornton CP, Rodney T, Thorpe RJ Jr, Allen J. Social cohesion in health: a concept analysis. *ANS Adv Nurs Sci*. 2020;43(4):375-390. doi:10.1097/ANS.0000000000000327
17. Wood SK. Transition to cancer survivorship: a concept analysis. *ANS Adv Nurs Sci*. 2018;41(2):145-160. doi:10.1097/ANS.0000000000000190
18. Rodgers BL. Concept analysis: an evolutionary view. In: Rodgers BL, Knafl KA, eds. *Concept Development in Nursing: Foundations, Techniques, and Applications*. W. B. Saunders; 2000:77-102.
19. Harper MG. Ethical multiculturalism: an evolutionary concept analysis. *ANS Adv Nurs Sci*. 2006;29(2):110-124. doi:10.1097/00012272-200604000-00004
20. Lynam DR, Caspi A, Moffit TE, Wikström PO, Loeber R, Novak S. The interaction between impulsivity and neighborhood context on offending: the effects of impulsivity are stronger in poorer neighborhoods. *J Abnorm Psychol*. 2000;109(4):563-574. doi:10.1037//0021-843x.109.4.563
21. Guerrieri R, Nederkoorn C, Jansen A. The interaction between impulsivity and a varied food environment: its influence on food intake and overweight. *Int J Obes Lond*. 2008;32(4):708-714. doi:10.1038/sj.ijo.0803770
22. Ogletree SM. Doing the right thing: determinism, moral responsibility and agency. *Int J Humanit Soc Sci*. 2013;3(13):1-7.
23. Papachristou H, Nederkoorn C, Havermans R, van der Horst M, Jansen A. Can't stop the craving: the effect of impulsivity on cue-elicited craving for alcohol in heavy and light social drinkers. *Psychopharmacol*. 2012;219(2):511-518. doi:10.1007/s00213-011-2240
24. Selvi Y, Aydin A, Atli A, Boysan M, Selvi F, Besiroglu L. Chronotype differences in suicidal behavior and impulsivity among suicide attempters. *Chronobiol Int*. 2011;28(2):170-175. doi:10.3109/07420528.2010.535938
25. Reed DD, Kaplan BA, Brewer AT. A tutorial on the use of Excel 2010 and Excel for Mac 2011 for conducting delay-discounting analyses. *J Appl Behav Anal*. 2012;45(2):375-386. doi:10.1901/jaba.2012.45-375
26. Paaver M, Eensoo D, Pulver A, Harro J. Adaptive and maladaptive impulsivity, platelet monoamine oxidase (MAO) activity and risk-admitting in different types of risky drivers. *Psychopharmacol*. 2006;186(1):32-40. doi:10.1007/s00213-006-0325-3
27. Doran N, Cook J, McChargue D, Spring B. Impulsivity and cigarette craving: differences across subtypes. *Psychopharmacol*. 2009;207(3):365-373. doi:10.1007/s00213-009-1661-x
28. Patros CH, Alderson RM, Kasper LJ, Tarle SJ, Lea SE, Hudec KL. Choice-impulsivity in children and adolescents with attention-deficit/hyperactivity disorder (ADHD): a meta-analytic review. *Clin Psychol Rev*. 2016;43:162-174. doi: 10.1016/j.cpr.2015.11.001
29. McClain MB, Mills AM, Murphy LE. Inattention and hyperactivity/impulsivity among children with attention-deficit/hyperactivity-disorder, autism spectrum disorder, and intellectual disability. *Res Dev Disabil*. 2017;70:175-184. doi:10.1016/j.ridd.2017.09.009
30. Stanford MS, Mathias CW, Dougherty DM, Lake SL, Anderson NE, Patton JH. Fifty years of the Barratt Impulsiveness Scale: an update and review. *Pers Individ Dif*. 2009;47(5):385-395. doi: 10.1016/j.paid.2009.04.008
31. Du W, Green L, Myerson J. Cross-cultural comparisons of discounting delayed and probabilistic rewards. *Psychol Rec*. 2002;52(4):479-492.
32. Niv S, Tuvblad C, Raine A, Wang P, Baker LA. Heritability and longitudinal stability of impulsivity in adolescence. *Behav Genet*. 2012;42(3):378-392. doi:10.1007/s10519-011-9518-6
33. Robinson MD, Pearce EA, Engel SG, Wonderlich SA. Cognitive control moderates relations between impulsivity and bulimic symptoms. *Cognit Ther Res*. 2009;33(4):356-367.
34. Brodsky BS, Oquendo M, Ellis SP, Haas GL, Malone KM, Mann JJ. The relationship of childhood abuse to impulsivity and suicidal behavior in adults with major depression. *Am J Psychiatry*. 2001;158(11):1871-1877. doi:10.1176/appi.ajp.158.11.1871

35. Chisholm JS. Attachment and time preference. *Hum Nat.* 1999;10(1):51-83. doi:10.1007/s12110-999-1001-1
36. Allen C, Nettle D. Hunger and socioeconomic background additively predict impulsivity in humans. *Curr Psychol.* 2019;5:1-5. doi:10.1007/s12144-019-0141-7
37. Metcalfe J, Mischel W. A hot/cool-system analysis of delay of gratification: dynamics of willpower. *Psychol Rev.* 1999;106(1):3-19. doi:10.1037/0033-295x.106.1.3
38. Daniel TO, Stanton CM, Epstein LH. The future is now: reducing impulsivity and energy intake using episodic future thinking. *Psychol Sci.* 2013;24(11):2339-2342. doi:10.1177/0956797613488780
39. Ryu DW, Kim JS, Yoo SW, Oh YS, Lee KS. The impact of impulsivity on quality of life in early drug-naïve Parkinson's disease patients. *J Mov Disord.* 2019;12(3):172-176. doi:10.14802/jmd.19004
40. Sokić K, Horvat D, Martinčić SG. The Role of Impulsivity in Post-Purchase Consumer Regret. In: Proceedings of the ENTRENOVA - ENTERprise REsearch InNOVation Conference, Rovinj, Croatia, September 12-14, 2019. <https://ssrn.com/abstract=3492256>
41. Lin SP, Shih HC, Huang YC. Emotional states before and after impulsivity. *Soc Behav Pers.* 2009;37(6):819-824. doi:10.2224/sbp.2009.37.6.819
42. Rung JM, Peck S, Hinnenkamp JE, Preston E, Madden GJ. Changing delay discounting and impulsive choice: implications for addictions, prevention, and human health. *Perspect Behav Sci.* 2019;42(3):397-417. doi:10.1007/s40614-019-00200-7
43. Reed DD, Naudé GP. Impulsivity and self-control. In: Hupp S, Jewell J, eds. *The Encyclopedia of Child and Adolescent Development.* Wiley: 2020;1-10. doi:10.1002/9781119171492.wecad086
44. Paasche C, Weibel S, Wittmann M, Lalanne L. Time perception and impulsivity: a proposed relationship in addictive disorders. *Neurosci Biobehav Rev.* 2019;106:182-201. doi:10.1016/j.neubiorev.2018.12.006
45. Skinner BF. *About Behaviorism.* Knopf; 1974.
46. Berkman ET, Graham AM, Fisher PA. Training self-control: a domain-general translational neuroscience approach. *Child Dev Perspect.* 2012;6(4):374-384. doi:10.1111/j.1750-8606.2012.00248.x
47. Martínez Loredó V, Fernández Hermida JR. Impulsivity-targeted selective preventive interventions and treatments in addictive behaviors. *Revista de Psicología Clínica con Niños y Adolescentes.* 2019;6(3):1-7.
48. McGovern MP, Carroll KM. Evidence-based practices for substance use disorders. *Psychiatr Clin North Am.* 2003;26(4):991-1010. doi:10.1016/s0193-953x(03)00073-x.
49. Braet C, Claus L, Verbeken S, Van Vlierberghe L. Impulsivity in overweight children. *Eur Child Adolesc Psychiatry.* 2007;16(8):473-483. doi:10.1007/s00787-007-0623-2